

Harsh Pal

📍 Bengaluru, India ✉ theharshpal2306@gmail.com ☎ +91 8368802064 📁 Portfolio in harrssh 🐙 GitHub

Profile Summary

Software Engineer experienced in building **scalable APIs**, **distributed systems**, and **enterprise-grade applications**, with hands-on exposure to **AI-powered multi-agent systems**. Targeting a **Back-end** or **Fullstack Engineering** role to deliver **high-performance**, secure software by leveraging deep expertise in **API development**, **microservices architecture**, **cloud-native technologies**, **LLM integration**, and **agentic AI workflows**.

Technical Skills

Languages: C++, C#, Java, Python

Technologies: .NET, Unity, Next.js, React.js, Node.js, Spring Boot, LLM Tool Calling, RAG

Concepts: Low-Level Design, High-Level Design, Concurrency & Multi-threading, Data Structures & Algorithms, OOP, Databases

Tools & Platforms: Git, CI/CD, PostgreSQL, Docker, Agentic AI Workflows, Kafka, Kubernetes, RabbitMQ

Experience

Software Engineer

WiseTech Global Pvt. Ltd.

Bengaluru, Karnataka

July 2024 – Present

- Built a **multi-agent AI orchestrator** to automate core developer workflows, utilizing an iterative **agent-debate mechanism pipeline** to reduce manual **task duration by 80%**.
- Optimized AWS cloud SSM API calls reducing by **60%** via caching/batch retrieval, minimizing attack surface & costs.
- Refactored **monolithic service layers** into **modular REST APIs**, leveraging **multi-threading** to parallelize complex calculation logic and reduce processing time by **35%**, while enforcing **SOLID** and **least-privilege** principles.
- Developed **enterprise-grade APIs** with **non-blocking I/O** and **async processing**, reducing average response latency by **40%** across **high-concurrency workloads** via robust **multi-threaded error-handling**.
- Strengthened **system security and resilience** by implementing a robust exception handling framework for critical errors reducing downtime by **40%**.
- Implemented **security change extenders** to enforce **write-lock constraints** on closed entities, blocking unauthorized backend modifications and reducing **data integrity violations by 60%** across critical workflows.
- Tuned **database query performance** and **API response handling**, achieving a **30% reduction in p95 latency** (5ms per call) and boosting throughput by **20%** for high-volume transactional workloads.
- Led the extraction of a production-critical service from a **monorepo** into a standalone repository while preserving full Git commit history, enabling independent **CI/CD pipelines** and **100% zero-downtime production cutover**.
- Led **design and code reviews** across a **5+ engineer** cross-functional team, improving **test coverage by 20%**, reducing **post-merge defects by 30%**, and enforcing **architectural standards**.
- Performed **security-aware root cause analysis** on production incidents through **log analysis**, **metric monitoring**, and **anomaly detection**, resolving **15+ critical issues** and reducing **mean time to resolve by 35%**.
- Automated CI/CD code quality checks with **Bash**, **pylint**, **mypy**, and **isort** to ensure **100%** consistent quality across teams.

Research Intern

University of Victoria

Victoria, BC, Canada

May 2023 – Aug 2023

- Published research in **BioResources Journal** on **micromechanical modeling of thermal conductivity** in wood composites, contributing predictive models to advance **sustainable structural material design**.
- Derived generalizable **thermal conductivity equations** for **5+ cross-laminated timber (CLT)** panel configurations, enabling accurate thermal property estimation for novel composite architectures **without physical testing**.

Projects

Curio.Ai - AI Powered Personalized Learning Platform [🔗](#) | *Java, Spring Boot, REST APIs, AI/ML*

- Developed an AI-enabled learning platform that generates personalized learning pathways by analyzing user behavior, preferences & learning patterns.
- Architected scalable back-end services and RESTful APIs using a modular design approach, enabling adaptive content recommendations and seamless future integrations.
- Designed system components to support personalized content delivery, user progress tracking & intelligent recommendation workflow.

AR Based Game [🔗](#) | *C#, Unity Game Engine, Visual Studio*

- Developed a 3D FPS Mobile Game in Unity, incorporating AR technology for enhanced player immersion.
- Utilized AR Foundation to seamlessly integrate virtual enemies into real-world environments through the device camera.

AI Enabled Blogging Platform [🔗](#) | *TypeScript, Node.js, Next.js, Jest*

- Built a full-stack blogging platform featuring secure authentication, content management, blog publishing workflow & a responsive UI.
- Developed scalable back-end services and intuitive front-end experiences to support content creation, editing, and administration.
- Implemented comprehensive unit and integration testing using Jest, improving application reliability, maintainability, and code quality.
- Incorporated modern software engineering practices, including modular architecture, reusable components, and secure user management.

Achievements

- Won **Professional Development & Innovation Award** – IIT Roorkee Convocation 2024
- **Globalink Research Fellow** – 12-week fully-funded research program in **Canada**
- **Expert** on [Codeforces](#)  & **Knight** on [LeetCode](#) ; solved 1,000+ problems across major competitive programming platforms.

Education

Indian Institute of Technology (IIT), Roorkee

Nov 2020 – July 2024

Bachelor of Technology in Mechanical Engineering — CGPA: 8.19